Chemistry 3A – Fall 2025 Tuesday Version**: Section 25810**

Name _____ ID _____

Mini-Midterm on Gases and Gas Law T**otal Points:** 28 Points (Extra Credit)

Instructions: Show all work for calculation problems to receive partial credit. Use proper significant figures where applicable. No notes or cards are allowed. You are allowed an approved calculator, Green & Yellow Sheets. You do not need scratch paper:put it on the sheet

1. (4 pts, 1 pt each) Match each mathematical formula or statement to its corresponding law by entering the correct letter in the blank.

Choices:

___ $\frac{V_1}{T_1} = \frac{V_2}{T_2}$
___ $PV = nRT$
____ "Equal volumes of all gases, at the same temperature and pressure, contain the same number of molecules."
____ $P_1V_1 = P_2V_2$

(a) Boyle's Law
(b) Charles' Law
(c) Gay-Lussac's Law
(d) Avogadro's Law
(e) Ideal Gas Law

2. Standard Temperature and Pressure (STP) (3 pts)

(a) What is the numerical value and unit for Standard Pressure used in our course? (1 pt)

(b) What is standard temperature in Celsius (°C)? (1 pt)

(c) Convert that standard temperature into Kelvin (K). (1 pt)

3. Charles's Law Calculation (5 pts)

A flexible balloon is filled to a **volume** of $\mathbf{3.50\ L}$ at room **temperature** ($\mathbf{25^\circ C}$). It is placed in a warm environment until its **volume** expands to $\mathbf{4.25\ L}$ at constant pressure. What is the **new temperature** of the balloon in **Celsius (**°C**)**? (Are intermediate conversions necessary!?)

4. Kinetic Molecular Theory (3 pts)
According to the Kinetic Molecular Theory (KMT) for an ideal gas, describe what happens to the energy of gas particles during their collisions with each other and the container walls.

5. Ideal Gas Law Calculation (5 pts)
Calculate the pressure (in atmospheres) exerted by 0.150 moles of carbon dioxide gas (CO_2) confined in a 2.50 L container at a temperature of 310 K. (Use $R = 0.08205 \frac{\text{L·atm}}{\text{mol·K}}$)

6. Partial Pressures and Mole Fraction (8 pts)
A gaseous mixture in a sealed container consists of $\mathbf{0.400}$ moles of nitrogen gas (N_2) and $\mathbf{0.100}$ moles of oxygen gas (O_2). The total pressure of the gas mixture is measured to be $\mathbf{1.25}$ atm.

(a) What is the total number of moles ($\boldsymbol{n}_{\text{tot}}$) of gas in the container? *(1 pt)*

(b) Determine the mole fraction ($\boldsymbol{\chi}$) of oxygen gas (O_2). *(2 pts)*

(c) Calculate the partial pressure ($\boldsymbol{P}_{O_2}$) of oxygen gas in the mixture. *(3 pts)*

(d) If the total pressure inside the container increases while keeping the relative amounts of each gas constant, what happens to the mole fraction of O_2? Briefly explain why. *(2 pts)*